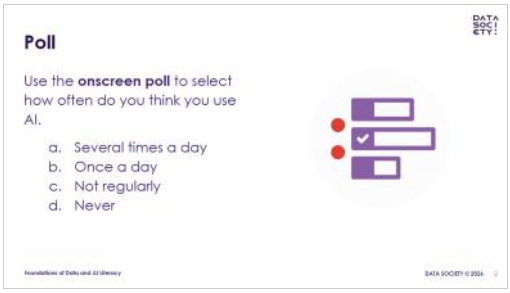
DAY 2 Exploring Data And AI Projects

Wednesday, 4 November 2026 07:07

DAY 2 Foundati...

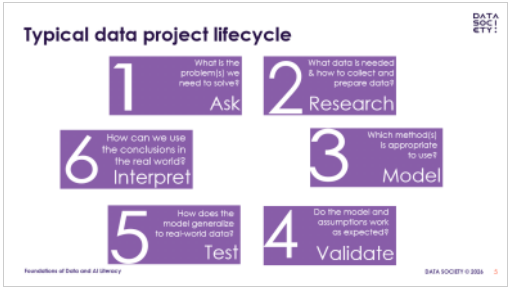




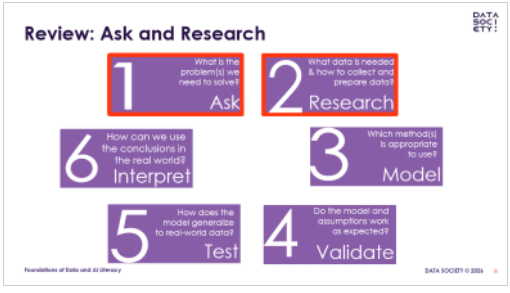


Joy also challenged the notion that at least one person said never. Although AI might not has been used in a LLM or general AI, they have probably used it indirectly. (eg. Alorithms in digital entertainment sources.)



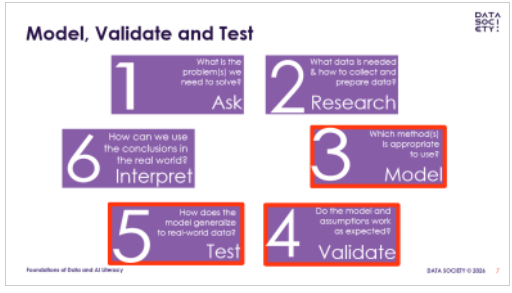


TECHNICAL STEPS



It is not always a problem, but something needs attention.

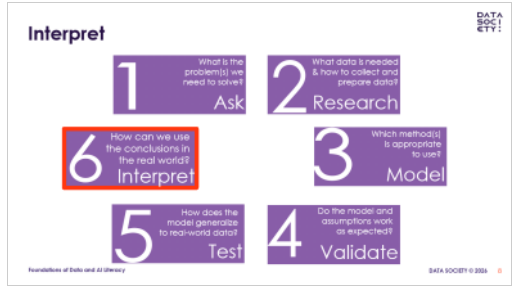
Sometimes the data that is available, or not available, will force you to go back and refine your question



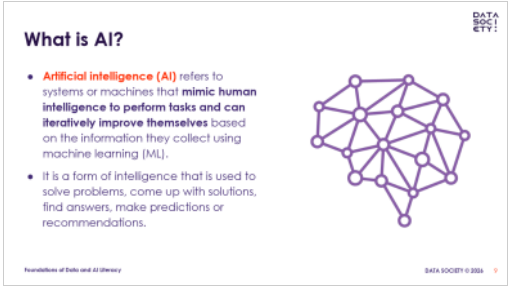
There are different methods and ways to sort your data (this is the model). Chose right.

Is all pieces of data relevant to this problem and ____

You need to train your model on data on which you know the answer and then justify that the model works correctly.

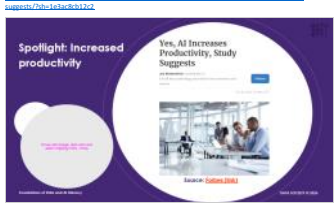


What do the results mean. Hoe do you interpret the prescriptions and predictions of AI



They improve with minimal human adjustment.

https://www.forbes.com/sites/joemckendrick/2023/04/25/yes-ai-increases-productivity-study-suggests/?sh=1e3ac8cb12c2



Ideally AI is not replacing jobs, it is replacing some jobs, but mostly helps and betters productivity of already existing jobs.

How AI works

- **To apply AI, you need data.** A lot of data!
- AI "algorithms" are trained using large datasets so that they can identify patterns, make predictions and recommend actions, much like a human would, just faster and better.
- There are several data science methods, including machine learning, applied to build AI technologies.



The raeson why AI has been taking of recently is due to the fact that it needed this much data, and a now there was a perfect spot where we have so much data that humans can't sift through it all, so computers were employed to help with this.

Note as well that so much data is being produced by AI as well that humans have it hard to go through all of the data produced by them as well.

AI examples in the finance sector

Enova uses AI in its lending system, **to provide analytics about finances and credit.**

Ocrolus uses AI and machine learning along with human verification **to speed up the analysis of financial documents.**

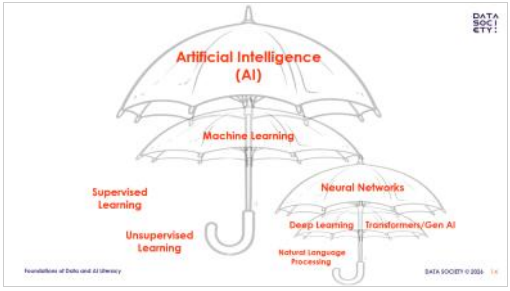
Socure uses AI and machine learning **to verify identity.**

Enova: Looking at applicants, finances and credits and comparing it to average mm. to see if they should be lending. It says nothing about if it is AI that makes the final decision, but it is part of.

The two other also sift through a lot of data to match a client to some other documents and standards.AI is not making decisions. It speeds up the process for a human.

A risk is the question of who is responsible for errors or if someone gets hurt

"if human is always in the loop why AI Bias and Hallucinations become big matter?"
If the AI has hallucinated or is bias. Then what you use to make decisions would give you incorrect decisions. So if the human only uses the AI for knowledge, it is extremely important to make sure that that is is as correct and objective as possible.

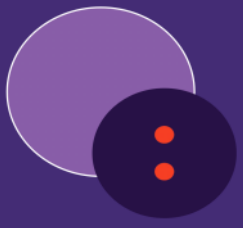


ML: Method that is applied to specific data tpo do one specific task. AI should be able to do multiple tasks over a broader space.

Why learn about these concepts?

To cultivate a mindset	To develop language	To enhance skills
Understand what options are available to optimize workflows, and drive success in your projects.	Gain basic technical understanding of AI systems and develop vocabulary and fluency in the "shared language".	Refine your communication and collaboration skills, especially when working with data science teams.

Machine learning methods



What is machine learning?

- **Machine learning (ML)** uses **algorithms** to find patterns in massive amounts of data and **predict** future results with minimal human intervention.
- It powers many of the services we use today:
  - Recommendation systems like those on Netflix
  - Search engines like Google
  - Voice assistants like Siri and Alexa
- Most is categorized as either supervised, unsupervised or semi-supervised.



It is almost like a math function. You take an input and get a output.

Eg. Spam filters: input is an email. The output is if the email should be labeled spam.

Machine learning: examples

Input	Algorithm (Set of steps to accomplish a task)	Output
Emails	Spam filtering functions	Email is a spam or not
Text in English	Machine translation functions	Text in French
Sequence of words	Functions to understand the meaning of each word and how they relate to each other	Recommend words to complete the sentence
Audio data	Speech recognition functions	Text transcripts
Stock market data (price, volume, etc.) over time	Predictive modeling algorithms	Forecast of future stock prices

Supervised learning

- An algorithm is given **tagged and labeled** input data to create such a solid map that we can predict the output for any novel input data.
- The goal is to map known input to a known output.
- For example, in predicting house prices, a human labels data and includes features like time of year, interest rates, square footage, etc. and specifies the output, i.e., the housing prices. The algorithm learns to identify patterns and correlations between the input and output and takes the features of a new house and predict its likely price.



We take a lot of data and separate the data so we can model and create the model on part of it, and then test it on the rest of the data.

Unsupervised learning

- An algorithm is allowed to detect and learn patterns based on **untagged** input data, usually to **generate categories.**
- The goal is to map known input to a unknown output.
- For example, by analyzing website customer journey data such as pages visited, clicks, etc., the algorithm autonomously identifies patterns and groups users with similar behavior.



We let the AI tell us or recognize if there are any patterns (tex customer segmentation, or sales spikes). It is patterns and specific things that you didn't realize was a pattern or problem.

Say we have a bookstore, Half of the books will be shelved by supervised learning as you tell it to sort the books by alphabetical order. Half of the books will be sorted by unsupervised learning and you ask them to sort the books. You do not know if they will sort it by size or title name or colour. But say that the bookstore sells books and board games. The supervised learning (if the data that teaches the model didn't include games) will sort the games in between the books based on creator name. But unsupervised learning might however separate books and games.

ML foundational methods families

- Machine learning provides **a set of methods for data analytics**, and projects often involve strategically applying multiple methods to address the challenges.

CLUSTERING	CLASSIFICATION	REGRESSION
generating labels from unlabeled data	applying labels to novel data points	predicting the influence of various factors

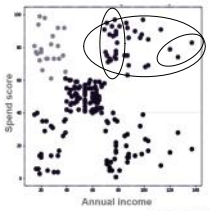
Clustering: is a way of combining data point to that from the cluster you can draw conclusions of what a new data point most likely is. Clustering is unsupervised. You don't know where or how many clusters there are. You just ask the model to identify the clusters.

Grouping data points into predefined groups. You start with a post or value and then you weigh it, is it part of this group or no. It usually qualitative. This is supervised learning.

Similar to classification, but uses numerical/quantitative data. You try to find a line that fits the data the bet &can be linnear or not). Where is the line that will predict the following values. This is suppervised

Clustering example

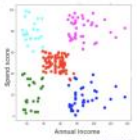
- **Clustering** is a way of grouping data points together based on their similarities and discovering hidden groupings in data.
- Since these groups **aren't labeled or predefined**, it requires domain knowledge to to interpret or extract insights.
- For example, a company uses clustering to identify previously unrecognized customer segments within their sales data and target advertising accordingly.
- Chat question: **Do you observe any customer segments emerging?**



A luxury car would be expected to be part of a cluster in the top right, but we do not know if it is a big group or small group or anything there between. When we get our clusters (say the colours below, they you can identify that this wuld be part of the purple class.

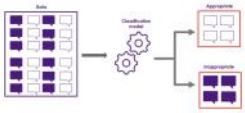
Clustering example (cont'd)

- Collaboration with domain experts helps define clusters, enabling clearer grouping, interpretation, and decision-making.
- The graph presented illustrates a potential visualization of the defined clusters.
- Chat question: **Which cluster would you market a new luxury car to?**



Classification example

- **Classification** is used for identifying and and grouping objects or ideas into **predetermined categories**.
- For example, LinkedIn uses a classification model to sort its posts as appropriate or inappropriate.
- The model is trained on a dataset of posts labeled as "appropriate" or "inappropriate" and sorts new posts based on similarity to the previously seen posts.



Regression example

- **Regression** helps **predict the value** of a variable based on the value of another variable(s).
- For example, when browsing homes online, you have likely encountered estimated property values on popular real estate platforms like Redfin and Zillow.
- These platforms leverage ML techniques like regression to predict the fair market value of homes.



ML advanced methods families

- Advanced machine learning methods build upon fundamental concepts to tackle complex problems and achieve higher accuracy and efficiency.

TEXT MINING	GRAPH ANALYSIS
inference and prediction based on textual data	finding patterns within a network to show connections

These are both semi-supervised as they can do both.

Text mining example

- **Text mining** helps analyze **unstructured text data** to extract meaning and identify patterns.
- For example, researchers used text mining to analyze tweets about different smartphone brands, like #iPhone and #Samsung, to understand consumer sentiment and trends.
- This allowed them to extract valuable information from the unstructured text data of Twitter.

Document info

ID	Title	Position	Key word
123	The Catcher in the Rye	6	feeling
334	Pride and Prejudice	4	behind
432	Brave New World	2	all
8854	Alice's Adventures in Wonderland	5	wonder
153	The Adventures of Huckleberry Finn	24	experience
44	To Kill a Mockingbird	7	sky
737	1984	1	willingly
847	Wuthering Heights	10	bio
790	Jane Eyre	12	society
938	The Lord of the Rings	3	near
8644	The Sun Also Rises	9	enemy

For example when looking for reviews. They can either look for keywords to get apicture of how the general look, but thus doesn't understand irony.

Sentimental looks at the whole context and then finds categories itself

Graph analysis example

- **Graphical analysis** uses visual representations like graphs and charts to **identify patterns, trends, and relationships between variables** within a dataset.
- For example, Google Shopping uses your searches and activity to identify related products you may be interested in.
- This it uses patterns and relationships to make personalized recommendations.



Image source: Kopp Online Marketing Consultant

AI methods

- There are three methods that play a key role in **building AI technologies**.

NEURAL NETWORKS	DEEP LEARNING	TRANSFORMERS
Modelled on the human brain, these algorithms train on large amount of data and have built in mechanisms to reduce modelling errors	Composed of neural networks that have more than 2 hidden layers. It is commonly used in text and image prediction	Instantly scans documents to map how every word relates to every other word in context

The problem om basing it on the brain is that there is a part that we don't understand and can't explain since we can't explain the brain itself. So we try to reduce error by moving around the knowledge we already know.

Deep learning: sometimes has more or less layers, but we know less about what is going on.

Transformers build on text and maps how every word relates to every other word in that document. (this is what models like ChatGpt is based on9

AI methods for computer vision

- **Computer vision**, a field of AI, applies neural networks and deep learning to understand and make decisions about images and videos.
- For example, in facial recognition systems, the computer sees only a grid of pixels, each with a specific brightness value.
- Here a neural network's task is to analyze the grid of numbers, decipher the patterns within, and ultimately identify the person in the image.



Source: Design and Dy Pineapple

This also needs to a trained on a lot of data.

AI project: Data requirements

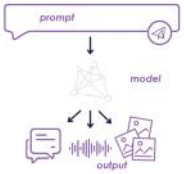
- Building AI requires **vast amounts of carefully curated data** that is:
  - Relevant real-world data
  - Properly classified/labeled data
  - Formatted data
  - Meets minimum volume requirement
- Additionally, pre-trained models and robust computing power can be helpful for efficient and effective training.



Image source: Created using generative AI tool

Generative AI

- **Generative AI** is a category of artificial intelligence algorithms that **produce new**, **realistic outputs** in the form of text, images, audio, video, or synthetic data.
- Typically, it starts with a simple prompt in which the user describes the output they want.
- Then, various algorithms generate new content according to what the prompt was asking for.



Multimodal AI Co-Pilots

- Modern AI systems are multimodal
- They accept and generate a variety of inputs and outputs


TEXT


IMAGE


AUDIO


VIDEO

Most models today (chat, Gemini, Claude) Are mainly based to create text. There are then other model that are specifically built to create images, audio, or video.

https://word.studio/gpt/

Instructor demo: Image generation

- Let's briefly experiment with an image generator.
- Submit **your image idea** that you would like to create in the chat (be creative!).



Note that sometimes multiple AI-chatbots will hallucinate the same thing. The might base it on the same data source or just find that the same vectors lay close to one or another, but they are not bullet proof.

When we did the Exercise: AI validation, and multiple chatbots (Gemini, ChatGPT an Claude) gave the 5 same sources. In some cases it gave different sources, this is most likely due to how you have used the chatbot in the past. It knows your preferences.

Key takeaways

- AI technologies are leveraged to drive automation, efficiency, innovation, and strategic advantage.
- Most data and AI projects combine a few methods to extract the full picture.
- The two big components that drive the decision for which method to use are: the **question** you are asking, and the **data** you have.



"The idea that you could poison your own algorithm with previous searches affection future results. The idea of not having a fresh perspective is concerning. Parochial.. Any thoughts on how to keep the familiar and the fresh search?"

Got no answer.

Summary

Today, we learned about:

- Data project lifecycle
- AI and its various subfields

Next session, we will learn about:

- Data management domains
- Tools and personnel




End of day 2
Foundations of Data and AI Literacy
DATA SOCIETY © 2026